

Assignment 5y

$$2. \text{ speed} = |v(t)|$$

$$(a) \text{ speed} = 2; \quad 2 \leq t \leq 4$$

$$\Rightarrow t = 3.12763$$

$$(b) s(t) = s(0) + \int_0^t v(x) dx$$

$$s(5) = 10 + \int_0^5 v(t) dt = -9.20733$$

$$(c) v(t) = 0, \Rightarrow t = \text{circled } 0.53603$$

$$t = 0.53603 \text{ and } 3.317756$$

$t=1$ is the only time between 0 and

4 where velocity is 0

$$(d) \text{ speed} = |v(t)|$$
$$\text{speed}' = (|v(t)|)' = \frac{v(t) \cdot v'(t)}{|v(t)|} = \frac{v(t) \cdot v'(t)}{|v(t)|}$$
$$= \frac{(-)(-)}{(+)} = +$$

The speed is increasing at time $t=4$
because $v(4) < 0$ and $v'(4) < 0$

$$(d) g'(x) = 3f(x) \cdot f'(x) \quad (\Rightarrow) g'(3) = 3 \cdot \frac{1}{4} \cdot 1 = \boxed{7.5}$$

4. (a) f has a local minimum where f' is $-$ and then $+$ and $f'(x) = 0$.
At $x = 6$ because $f'(x) = 0$ and f' is negative to the left of $x = 6$ and f' is positive to the right of $x = 6$.

(b) candidates: 0, 6, 8

~~$$f(1) - f(0) = \int_0^1 f'(x) dx$$

$$f(0) = f(1) - \int_0^1 f'(x) dx$$

$$f(0) =$$~~

~~$$f(8) - f(0) = \int_0^8 f'(x) dx$$~~

~~$$\Rightarrow f(0) = f(8) - \int_0^8 f'(x) dx$$~~

~~$$\Rightarrow f(0) = 4 - 12 = \boxed{-8} \leftarrow f(0)$$~~

~~$f(8) - f(6) = \int_6^8 f'(x) dx$~~

~~$$f(8) - f(6) = \int_6^8 f'(x) dx$$~~

~~$$\Rightarrow f(6) = f(8) - \int_6^8 f'(x) dx$$~~

~~$$\Rightarrow f(6) = 4 - 7 = \boxed{-3} \leftarrow f(6)$$~~

~~$$f(8) - f(0) = \int_0^8 f'(x) dx$$~~

~~$$\Rightarrow f(8) = -8 + 12 = \boxed{4} \leftarrow f(8)$$~~

~~$$\boxed{-8}$$~~

(c) f is concave down and increasing when $f'' < 0$ and $f' > 0$

And $f'' < 0$ when f' is $\downarrow$
So f is Concave Down and increasing over $(0, 1)$, $(3, 4)$